GORAP KUMAR

Cottbus, Germany (willing to relocate) | gorapkumarofficial@gmail.com
gorap50.github.io | linkedin.com/in/gorap-kumar-da | github.com/gorap50

PROFILE

M.Sc. Artificial Intelligence student at BTU Cottbus–Senftenberg working across **computer vision, machine learning and LLM applications**, with one year of industry experience on AI training data for healthcare and agriculture models, including leading a team of 10+ annotators. Hands-on with the full ML workflow: data preparation, model training and comparison, evaluation and edge deployment; currently building a RAG chatbot on a local LLM. Seeking a **working student (Werkstudent) position or internship** in AI / Machine Learning / Data Science; a mandatory internship (Pflichtpraktikum, min. 260 hours) is part of my degree. Available **immediately**; open to relocation within Germany. Interested in continuing with an industry Master's thesis.

EDUCATION

BTU Cottbus–Senftenberg **Cottbus, Germany**
M.Sc. Artificial Intelligence (Research Profile) — expected 2027 *Oct. 2024 – Present*

- Strong results: Introduction to Neural Signal Analysis (Grade: 1.0), Introduction to Computational Neuroscience (1.7)
- Further modules: Brain–Computer Interfaces for Neuroadaptive Technology, Modeling of Perception and Action, Biological Neuronal Networks, Learning in Real and Virtual Humans, Causal Data Science, Laboratory & Seminar
- Current semester: Data Analytics and Process Modelling

Sukkur IBA University **Sukkur, Pakistan**
B.Sc. Computer Science — CGPA 3.22/4.00 *Aug. 2019 – Aug. 2023*

- Key coursework: Machine Learning (B+), Artificial Intelligence (B+), Data Science (B+), Database Systems (Lab: A-), Software Engineering (B+), Operating Systems (Lab: A, Theory: A-), Parallel & Distributed Computing (A-), Cyber Security (A-), Compiler Construction (A-), Data Structures (B+)
- Final Year Project graded A | Sindh Endowment Scholarship (2022), Sindh Talent Hunt Program scholarship (2019)

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn), SQL, C++, JavaScript, Bash, R (DataCamp certified), Java, C# (coursework)
AI & Machine Learning: PyTorch, TensorFlow, Keras, Deep Learning, CNNs, Transfer Learning (MobileNetV2, ResNet), NLP, Random Forest, XGBoost, Class Imbalance (SMOTE), Model Evaluation & Error Analysis
Computer Vision & Robotics: OpenCV, YOLO, Image Classification & Object Detection, Medical Image Analysis, Explainable AI (Grad-CAM), Dataset Construction & Annotation (Darwin V7), ROS, Real-Time Edge Deployment (Raspberry Pi)
Generative AI & LLMs: Retrieval-Augmented Generation (LangChain, FAISS), Embeddings & Vector Search, Local LLMs (Ollama), Prompt Engineering, Streamlit, GenAI Dev Tools (GitHub Copilot, Claude, GPT)
Data Engineering: ETL Pipelines, Data Validation & Quality Assurance, Data Modelling, PostgreSQL, MongoDB
Web & Software: React, HTML/CSS, REST APIs (JSON), Backend & Database Development, Full-Stack Web & Mobile Apps
Infrastructure & Tools: Docker, Linux, Git/GitHub, CI Workflows, AWS, Jupyter, Google Colab, Power BI, Tableau, MS Office 365
Methods: Agile/Scrum (Jira), Confluence Documentation, Requirements Gathering, Team Leadership (10+ members), Workshop Facilitation
Languages: English (C1, fluent), German (A1–A2, actively learning), Sindhi (native), Urdu (fluent)

PROFESSIONAL EXPERIENCE

DataVision Pvt Ltd **Lahore, Pakistan (Remote)**
Data Annotation Engineer → Data Annotation Team Lead (from Jan. 2024) *Aug. 2023 – Aug. 2024*

- Led a team of 10+ annotators on large-scale healthcare and agriculture AI datasets, improving delivery speed by about 20% by finding and fixing workflow bottlenecks
- Annotated medical and precision-agriculture images at pixel level (Darwin V7) and managed the annotation workflow from task setup to final quality review
- Built Python/SQL validation scripts and data quality checks that reduced labeling inconsistencies by about 15% and saved roughly 5 hours per week
- Worked with ML engineers on error analysis of training data to find labeling issues that were hurting model performance
- Coordinated Data Science and Project Management teams with Jira and Confluence to shorten feedback loops on training data

PROJECTS

DocuMind — RAG Chatbot for PDFs (in progress) | *Python, LangChain, FAISS, Ollama, Streamlit* **2026**

– Building a chatbot that answers questions about uploaded PDFs using retrieval-augmented generation on a fully local LLM: document chunking, embeddings, FAISS vector search and source-grounded answers; no data leaves the machine

Glaucoma Detection Pipeline (HYGD) | *Python, TensorFlow/Keras, MobileNetV2, OpenCV* **2026**

– Benchmarked a custom CNN (92.6% val. accuracy) against MobileNetV2 transfer learning (98.7% val. accuracy) on clinical retinal fundus images
– Built a clinical-style test dashboard showing per-scan prediction, confidence and mismatch flags; evaluation focused on false negatives

Brain Tumor MRI Classification | *Python, TensorFlow/Keras, CNN* **2025**

– Built a 3-block CNN with dropout, data augmentation and early stopping for tumor / no-tumor classification on a small MRI dataset (253 images); 78.4% test accuracy

Spatial Computer Vision Microservices | *Python, Docker, Linux, Git* **2025**

– Containerized real-time spatial vision services in Docker, cutting processing latency by 20%; authored reproducible Bash deployment scripts and managed GitHub repositories with automated CI

Autonomous Weed Detection Robot (Edge AI) | *Python, YOLOv4, OpenCV DNN, Raspberry Pi, ROS* **2023**

– Deployed YOLOv4 with OpenCV DNN on a Raspberry Pi 4 for real-time crop/weed detection on a live camera feed, tuning input size and confidence/NMS thresholds for edge hardware in changing outdoor conditions

University Services Platform — Web Admin Portal & Mobile App | *React, JavaScript, Real-Time Database* **2023**

– Final Year Project (graded A): full-stack platform giving students one place for exam and class schedules, exchange programmes, scholarships and jobs, plus live university bus tracking; built the admin portal (React frontend, backend, database) independently

Maternal Health Risk Prediction | *Python, Scikit-learn, XGBoost, SMOTE* **2022**

– Compared six classifiers for prenatal risk prediction; Random Forest reached ∼85% accuracy after SMOTE balancing and feature pruning; automated the data pipeline, reducing manual data entry by ∼20 hours per week

INTERNATIONAL PROGRAMS & LEADERSHIP

EUNICE Intensive Program: Network & Cyber Security (3 ECTS) **Poznań, Poland**
Academic Participant, Poznan University of Technology *July 2026*

– Intensive European mobility program on secure network infrastructure, system vulnerabilities and threat analysis

Erasmus+ Training Course: "Skills Swap" **Dolný Ohaj, Slovakia**
Workshop Facilitator & Germany Team Representative (YEAH gGmbH) *June 2026*

– Designed and facilitated a workshop for 28 youth workers from 7 countries; built GitHub-hosted interactive quizzes for the closing session

CERTIFICATIONS

IMPACT IMPK – Cybersecurity Path (9 modules), Poznan University of Technology, 2026 | EUNICE Summer School in Network Security, 2026
DeepLearning.AI: Machine Learning Specialization | IBM: Python for Data Science, AI & Development
DataCamp: SQL Database Fundamentals, R Programming | Kaggle: Data Visualization